

5(a)	presence of sucrose (in sieve tube) lowers water potential (of phloem sap) ; A makes water potential more negative water enters (sieve tube), by osmosis / down water potential gradient ; increases volume (in sieve tube) ; increases <u>hydrostatic</u> pressure (in sieve tube at source) ; A turgor pressure <i>ref. to lower <u>hydrostatic</u> pressure (in sieve tube at sink) ;</i> <i>allow ecf for no ref. to, hydrostatic / turgor</i> detail ; e.g. sucrose removed at sink water follows sucrose that exits sink movement of, (phloem) sap / sucrose, down pressure gradient / from high to low hydrostatic pressure ; <u>mass flow</u> ;	4
5(b)(i)	<u>glycosidic</u> ;	1

Question	Answer	Marks
5(b)(ii)	nucleotide components are base, pentose sugar, phosphate ; UDP has, uracil and ribose and two phosphates / one extra phosphate ;	2
5(b)(iii)	<i>max three if describing breakdown of sucrose</i> <i>allow points from diagrams</i> <i>any four from</i> <i>induced fit</i> shape of, substrates / UDP and glucose, not (fully) complementary to shape of active site (of sucrose synthase) ; <i>lock and key</i> shape of, substrates / UDP and glucose, complementary to shape of active site (of sucrose synthase) ; <i>allow ecf if no ref. to shape</i> <i>induced fit</i> active site flexible / moulds around, substrates / UDP and glucose ; AW <i>lock and key</i> active site does not change shape / is not flexible / AW ; <i>induced fit</i> (active site moulds around so) <i>idea of</i> provides better fit / fully complementary ; <i>lock and key</i> substrate fits into active site ;	4
5(c)	<u>α-glucose</u> ; A glucose R β -glucose α , 1-4 ; α , 1-6 ; <i>bond types either way round</i> glycogen ;	4